D72e — The gravitational coupling as an emergent expression of the boundary leakage

Dependence $G(\alpha)$, the non-circularity architecture, and the registry of the twenty-three closed doors (verification campaign v25–v54)

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31 October 2026

Abstract

We present the consolidated synthesis of verification campaign v25–v54 of the PDL framework, which establishes the gravitational coupling G and the fine-structure constant α as two expressions of a single mean boundary-leakage parameter ε, propagated through the eighteen rungs of the declared structure by the cascade $\alpha_G = [\alpha(1+\delta)]^{18}$. The circularity affecting earlier determinations is broken *by architecture*: the predictive block is sealed from every measured input of G; its predictions are frozen by cryptographic fingerprint before any comparison; and the CODATA value of G is admitted only as an external arbiter. The frozen prediction $G_{\mathrm{pred}} = 6.6743264 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$ agrees with the CODATA 2018 arbiter at $+3.96$ ppm, i.e. $0.18\,\sigma$ of the experimental uncertainty. Twenty-three doors, each closed under declared multiplicity controls, are registered. The synthesis archives its own limits with equal care: a residual $\sim$20% fortuitous-agreement probability while the pre-registered falsifier ($d_r = 0.04795969 \pm 7 \times 10^{-8}$ simultaneously with $f = 0.157126$) remains open; and the identification of the missing radial embedding of the 918-site graph as the canonical obstruction to a final proof. Document lineage: D72 (founding interface asymmetry), D72b (short synthesis), D72d (long synthesis), D72e (this corpus document, prepared for Zenodo deposition).

1 Introduction: two constants, one leakage

Within the PDL framework the microscopic dynamics at the sea–boundary interface is far too complex to be modelled by a unique relation; the twenty-three closed doors of campaign v25–v54 are the negative trace of that complexity. The burden of the derivation is instead carried by a *mean* leakage behaviour: a single effective parameter ε per rung, repeated over the eighteen rungs of the declared structure. The electromagnetic coupling α (already derived structurally as $\alpha^{-1} = 137 + 33/917$, at $+0.09$ ppm of CODATA) and the gravitational coupling are then no longer two parallel constants: they are linked by

$$\alpha_G \;=\; \varepsilon^{18}, \qquad \varepsilon = \alpha\,(1+\delta), \qquad \delta \equiv \frac{\varepsilon}{\alpha} - 1, \tag{1}$$

and the gravitational constant follows from the declared conversion $\alpha_G = G\,m_p^2/(\hbar c)$. The small quantity δ is the asymmetry of the boundary flux sharing between the emitted channel (α) and the transmitted channel (ε). The task of the campaign was thus reduced to two numbers, ε and δ — and to proving that they are predicted, not fitted.

2 The derivation chain

The sealed chain used for the prediction is

$$\alpha^{-1} = 137 + \tfrac{33}{917} \tag{2}$$

$$\delta_{\text{pred}} = (1-x)\,\underbrace{2\big(1-\cos\tfrac{\pi}{18}\big)}_{\text{chord closure}} + x\,\underbrace{\Big(\tfrac{\pi}{18}\Big)^2}_{\text{arc closure}} = 3.0429555\,\% \tag{3}$$

$$\varepsilon_{\text{pred}} = \alpha(1+\delta_{\text{pred}}) = 0.0075194084307 \tag{4}$$

$$\alpha_{G,\text{pred}} = \varepsilon_{\text{pred}}^{18} = 5.906173 \times 10^{-39} \tag{5}$$

$$G_{\text{pred}} = \alpha_{G,\text{pred}}\,\frac{\hbar c}{m_p^2} = 6.6743264 \times 10^{-11}\ \text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}. \tag{6}$$

The target value $\delta = 3.0430\,\%$ is *bracketed* by the two closures: the chord closure alone undershoots it ($-1\,476$ ppm), the arc closure alone overshoots it ($+1\,062$ ppm); the surviving mixture is $x = 7/12$.

3 The three geometric routes to $x = 7/12$

The mixing fraction is not a fit: it is produced independently by three a priori geometric routes from the declared pieces of the structure,

$$x = \frac{24 - 180/18}{24} = \frac{14}{24}, \qquad x = \frac{\text{spf}(917)}{12} = \frac{7}{12}, \qquad x = \frac{\text{gpf}(133) - 12}{12} = \frac{7}{12}, \tag{7}$$

with $917 = 7 \times 131$ (the denominator of the structural α), $133 = 7 \times 19$ (the support of the historical champion $1/133$) and $180/18 = 10°$ the natural closure phase of the eighteen-rung structure. Three independent derivations converge on the same value: $7/12$ emerges from the geometry. In full honesty, its status is archived as a *reading* — a structural derivation with remarkable convergence, not yet a compelled deduction — pending the boundary operator of Section 6.

4 The non-circularity architecture

The decisive methodological result of the campaign is that circularity was broken by construction, not by argument. **(i) Sealed predictive block.** Permitted inputs: the declared structure (18 rungs, 918 sites, 24-regular shell, boundary/sea decompositions $11\,017 = 23 \times 479$, $10\,087 = 7 \times 11 \times 131$, 930), the exact constants h and c, and the measured proton mass m_p. Forbidden inputs: G_{CODATA}, the measured α_G, and the anchored values $\varepsilon_{\text{rung}}$, δ_{rung}. **(ii) Frozen predictions.** All outputs were frozen by SHA-256 fingerprint before any comparison (v53: `95bc1c472c21d100`; v54: `b3a697e5c75af550`). **(iii) External arbiter.** Only after freezing was the CODATA 2018 value $G = 6.67430 \times 10^{-11}$ admitted, with its 22 ppm uncertainty. Verdict:

$$\frac{G_{\text{pred}} - G_{\text{CODATA}}}{G_{\text{CODATA}}} = +3.96\ \text{ppm} = 0.18\,\sigma. \tag{8}$$

The role of the measured G is thereby inverted: it is an output predicted by the structure, not an input anchoring the structure.

5 Registry of the twenty-three closed doors

Campaign v25–v54 closed twenty-three doors, each under declared a priori membership and random-expectation (multiplicity) controls. Doors v25–v41 (closed-form and denominator fam-

ilies) are archived individually in the artefact manifest shipped with this package (`scripts/CAMPAIGN_ARTEFACTS.md`); doors v42–v54 are summarised below.

Version	Scope declared a priori	Verdict
v42	Full-boundary numerator $\delta \approx 11017/362043 = (1/33)(479/477)$	+22.6 ppm; structural, outside ±7 ppm
v43	Continuity controls (1/133, 335/11016)	closed
v44	$\varepsilon = L/S$ with free fractional L; 36 declared (L, S) pairs	hits(7 ppm) = 0
v45	"Total minus one" decoherent-link denominators	no restoration within ±7 ppm
v46	Denominator band scan $D \in [500, 10\,087]$, free numerator	chance level
v47	Unconstrained best rationals (Dirichlet regime)	ppb agreements, no structural factorisation
v48	Four closed analytic routes (topological, gambler's ruin, evanescent, T_{13})	support 133 required, not derived
v49	A priori generators of 133 (arithmetic families, harmonics, shell modes)	multiplicity ≈ controls
v50	Dephased two-channel flux; closure phase $180°/18 = 10°$	δ off by $\sim 1\,500$ ppm
v51	Damped closure refinement	chord/arc bracket; anchoring signature
v52	A priori geometry of $x = 7/12$; window-intersection test	empty at 7 ppm (anchoring theorem)
v53	Circularity break by architecture; frozen prediction	G_{pred} at +3.96 ppm ($0.18\,\sigma$)
v54	Boundary operator for the falsifier d_r AND f	not constructible; radial embedding missing

Certificates (scripts, JSON summaries, reports, consoles) of every door are published; the surviving structural readings are: $\varepsilon \sim 1/133$ at −81.5 ppm (sixteen consecutive re-evaluations, outside the window), 77/10240 at +16.6 ppm, and (1/33)(479/477) at +7.4/+21.9 ppm.

6 Honest caveats

The credibility of this document rests on stating its own limits as sharply as its result. **(a)** An agreement at $0.18\,\sigma$ carries a ∼20% probability of being fortuitous; the architecture removes anchoring, not chance. **(b)** The pre-registered falsifier — the simultaneous restoration of $d_r = 0.04795969 \pm 7 \times 10^{-8}$ and $f = 0.157126$ under the single generator $x = 7/12$ — is *not* closed: best structural approaches are $f \approx 1731/11017$ (−33 ppm; tolerance 7 ppm) and $d_r \approx 44/918$ (−613 ppm; tolerance 1.46 ppm). **(c)** The window-intersection theorem (v52) shows that the 22 ppm measurement noise of G forbids dual-anchor validation at ±7 ppm; final validation must proceed through the falsifier alone. **(d)** The missing canonical component is identified surgically: a radial embedding of the 918-site graph, required to construct the boundary operator that produces d_r and f from the same law.

7 Reproducibility and artefacts

This package contains: the present source (`D72e.tex`) with its bibliography (`refs.bib`) and compiled document (`D72e.pdf`); the consolidated sealed-pipeline script (`scripts/D72e_pipeline_v55.py`) reproducing the derivation chain of Section 2 together with its external-arbiter verdict, and its frozen output (`scripts/D72e_pipeline_v55_output.json`); and the campaign artefact manifest (`scripts/CAMPAIGN_ARTEFACTS.md`) listing the published certificates of every door v25–v54. The framework repository is `https://github.com/laubscher-lab/PDL-framework`. Suggested deposition licence: CC BY 4.0.

8 Conclusion

The campaign establishes the dependence $G(\alpha)$ at the level of the mean leakage field: both constants are expressions of the single parameter ε through the cascade $\alpha_G = [\alpha(1+\delta)]^{18}$. The circularity of G is broken by architecture — a sealed predictive block, fingerprint-frozen predictions, and G_{CODATA} demoted to the role of external arbiter, against which the frozen prediction stands at $0.18\,\sigma$. The registry of the twenty-three closed doors demonstrates that the surviving result is not a coincidence harvest but the last form standing. Final proof awaits the boundary operator able to restore d_r and f simultaneously; the radial embedding of the 918-site graph is the identified path to it.

References

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